Q.29 What are the advantages and limitations of a Flexible Manufacturing System (FMS)?

Q.30 What is a robot? Describe the different robot motions and configurations.

Q.31 Explain the process of transferring a drawing from CAD software to CAM and vice versa.

Q.32 Explain the purpose of fillets and chamfers in solid modeling?

Q.33 What is a tool path? Explain the different parameters specified during tool path generation.

Q.34 Explain the concept of different terms used in CNC operations like Clearance, Retract, Feed plane and Depth of cut.

Q.35 What are the key features of a Flexible Manufacturing system (FMS)?

SECTION-D

Note: Long answer type questions. Attempt any two questions out of three questions. (2x10=20)

Q.36 Explain the product cycle in the context of CAD/CAM automation. Discuss the reasons for implementing CAD/CAM.

Q.37 Describe the process of creating a composite solid using Boolean functions and then performing 3D operations like array, mirror and rotate.

Q.38 Explain the concept of Flexible Manufacturing Systems (FMS). Discuss the principles of flexibility and the different features of FMS?

(20)

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6th Sem / Mech, Mecatronics
Sub : CAD/CAM/CAD CAM & FMS

Time : 3Hrs. M.M. : 100

SECTION-A

Note: Multiple choice questions. All questions are compulsory (10x1=10)

Q.1 Which of the following is not a solid primitive?
a) Box b) Cone
c) Extrude d) Wedge

Q.2 What is a common operation in turning machines?
a) Pocketing b) Facing
c) Contouring d) Facing

Q.3 Which of the following is used for creating a composite solid?
a) Array b) Union
c) Mirror d) Rotate

Q.4 Which of the following is a feature of a Flexible Manufacturing system (FMS)?
a) Production equipment
b) Robot configuration
c) Robot sensors
d) Robot motions

Q.5 What is the process of converting the generated tool path into NC code called?
a) Back plotting b) Post processing
c) Tool path generation d) Tool chaining

Q.6 What is the process of setting up parameters for the transfer of programs to a CNC mcahine called?

a) Communication
b) Post-Processing
c) Back plotting
d) Chaining the geometry

Q.7 Which of these is a command for modifying solid edges and faces?

a) Shell b) Fillets
c) Extrude d) Revolved

Q.8 Which of the following is a milling operation?

a) Threading b) Grooving
c) Pocketing d) Parting

Q.9 Which of the following is a command for 3D operations?

a) Shell b) Fillets
c) 3D array d) Revolved

Q.10 What is a key step involved in a CAM operation?

a) Creation of fillets
b) Dimensioning of solids
c) Introduction of CAD/CAM
d) Post-processing

SECTION-B

Note: Objective type questions. All questions are compulsory. (10x1=10)

Q.11 What are the three Boolean functions used for creating a composite solid?

Q.12 Name few Robot application.

Q.13 Define CAD, CAM.

Q.14 Define F.M.S.

Q.15 Name any two tool materials used in CNC?

Q.16 List two advantages of CAD?

Q.17 What is the purpose of back plotting and verification of an operation?

Q.18 Name a command used converting a 2D drawing into 3d drawing.

Q.19 What are the two types of flexibility in a manufacturing system?

Q.20 What are the two types of spaces in which a drawing can be arranged?

SECTION-C

Note: Short answer type questions. Attempt any twelve questions out of fifteen questions. (12x5=60)

Q.21 Explain the concept of solid modeling and list the different solid primitives available in AutoCAD.

Q.22 Describe the steps involved in a CAM operation.

Q.23 Explain the different Boolean functions (Union, Subtraction and Intersection) with suitable examples.

Q.24 What are the different operations involved in milling? Explain any three.

Q.25 What are the different operations involved in turning? Explain any three.

Q.26 Explain the concept of SW, SE, NE and Isometric views in 3D space.

Q.27 Describe the steps involved in the creation of a composite solid using Boolean functions.

Q.28 Explain how you would set up a job for tool path generation and verification.